Multi-Module Simultaneous ASSC Adequacy

DBA-MA-SMR-FC1-TB1

Thermal-Hydraulic Adequacy of the 72-Hour Coping Floor Under Simultaneous Multi-Module Actuation — A Technical-Basis Supplement to DBA-MA-SMR-FC1

A Facility-Class Technical-Basis Supplement of the DBA-MA Nuclear Series · Parent: DBA-MA-SMR-FC1

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Closes the calibration item registered at DBA-MA-SMR-FC1 §9.1 and substantiates the coping-duration barrier credited at FC1 §5A and treated as the governing "clock" cutset at DBA-MA-DiD §5. Quantitative inputs are design-certification data, carried as calibration items.

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1. Purpose, Scope, and Position in the Series

1.1 Purpose

DBA-MA-SMR-FC1 credits a 72-hour Autonomous Safe Shutdown Condition (ASSC) design-basis duration — the period for which the near-term integral-LWR SMR sustains safe shutdown and decay-heat removal without operator action, external power, or active makeup. That figure is derived, in the design-certification basis, from single-module analysis. FC1 §5A.4 declares as a residual, and FC1 §9.1 registers as an open calibration item, the question this supplement answers: does the 72-hour coping duration hold when every module of a multi-module campus actuates its passive systems simultaneously, as a correlated common-cause or adversarial event would drive them to do? This document supplies the thermal-hydraulic adequacy framework for that question — the governing physics, the decomposition that locates where the multi-module concern actually lives, the conservative bounding analysis, the acceptance criterion, and the register of certified inputs that populate the proof for a specific design.

The document deliberately does not assert numeric coping durations, pool inventories, or decay-heat values. Those are design-certification quantities specific to each design family, and inventing them would defeat the purpose of a technical basis. The contribution here is the analysis structure and the two findings that structure produces: that multi-module adequacy reduces almost entirely to the sizing basis of the shared ultimate heat sink, and that the 72-hour class floor must be reconciled against the family-specific certified coping durations, which the multi-module case is precisely what tests.

1.2 What This Instrument Is, and Is Not

This is a facility-class technical-basis supplement. It develops one analysis — the thermal-hydraulic adequacy of the ASSC coping duration under simultaneous multi-module actuation — and its result is cited by the facility-class design basis rather than re-derived there. It is not a source-term analysis (that is FC1 §3A and DBA-MA-ST), not an adversarial design basis (FC1 §4A), and not a consequence chain (the use-case instruments). It answers exactly one question, and it answers it as a bounding, deterministic adequacy demonstration in the established safety-analysis idiom, so that a design-certification reviewer can populate it with certified inputs and either confirm the 72-hour floor for a campus or identify the specific shared resource that fails to support it.

1.3 Position in the Hierarchy

Parent. DBA-MA-SMR-FC1 (the facility-class design basis). This supplement is a child of the class document, in the same relationship DBA-MA-ST bears to FC1's source-term treatment: the derivation lives here; the design basis states the result and references it.

Inherited (referenced by owner). The ASSC construct and the 72-hour design-basis duration (FC1 §5A.1); the multi-module campus configuration and its shared-infrastructure inventory (FC1 §2.2); the passive-safety adequacy result and its declared residual (FC1 §5A.4); the multi-module inventory-aggregation observation (FC1 §3A.5); and the source-term conditionality principle it rests on (DBA-MA-ST, via FC1 §3A.6). The barrier-correlation frame is DBA-MA-DiD, which treats the coping clock as a governing cutset; this supplement is the physical substantiation of that cutset.

Locally defined. The thermal-hydraulic coping heat balance as applied to the multi-module campus; the ultimate-heat-sink architecture taxonomy; the adequacy decomposition into per-module-independent and shared-resource elements; the conservative bounding assumptions and the acceptance criterion; and the calibration-input register. The result produced by this supplement populates FC1 §5A.1 and closes FC1 §9.1 on FC1's next revision pass.

1.4 The Question, Stated Precisely

A campus of N near-term integral-LWR SMR modules experiences a correlated event — loss of the shared switchyard, loss of shared cooling-water supply, or a cyber-kinetic combined action — that drives all N modules to actuate passive safety at effectively the same time. Each module's coping clock starts together and, absent intervention, would expire together. The single-module design basis establishes that one module copes for at least the ASSC duration without operator action, external power, or makeup. The precise question is whether the campus as a whole — drawing on whatever ultimate-heat-sink inventory, makeup provisions, and heat-rejection paths its modules share — sustains every module within its coping design limits for the full ASSC duration when all N clocks run at once. The answer is not assumed; it is the object of the proof.

2. The Governing Physics of Passive Coping

2.1 Decay Heat and the Coping Heat Balance

After reactor shutdown, fission ceases, but the core continues to generate decay heat from the radioactive decay of fission products and actinides. The decay-heat fraction is a strong, decreasing function of time after shutdown: on the order of six to seven percent of rated thermal power at the instant of shutdown, falling to roughly one to one-and-a-half percent within an hour, and to a fraction of one percent within a day, following the standard decay-heat characterization (ANS-5.1) evaluated at the core's burnup and operating history. Because the fraction is front-loaded, the integrated energy the passive system must reject over a coping window of tens of hours is dominated by the first several hours, and the design-basis coping analysis must use the bounding decay-heat curve — end-of-cycle, peak burnup — rather than a nominal one.

Passive coping without makeup is, at its core, a heat-balance argument. The passive system rejects decay heat to a water inventory — a reactor pool, an isolation-condenser pool, or a passive containment cooling reservoir. That inventory first absorbs sensible heat as it warms to saturation, and then absorbs the far larger latent heat of vaporization as it boils. The coping duration without makeup is set by the time required to consume the usable inventory: coping ends, in the limiting sense, when the integrated decay-heat energy delivered to the sink equals the usable inventory multiplied by the sum of its sensible and latent heat capacities. Expressed for a campus of N modules, the coping duration is defined implicitly by the balance.

$$\int_0^{t_{cope}} \sum_{i=1}^{N} P_{decay,\,i}(t)\ dt \;=\; M_{usable}(c_p\,\Delta T_{sub} + h_{fg})$$

where the left side is the decay-heat energy delivered to the shared sink over the coping window and the right side is the energy the sink can absorb: t_{cope} is the coping duration; $P_{decay,i}(t)$ is the decay heat of module i, following the ANS-5.1 characterization at that module's burnup and operating history; M_{usable} is the usable shared inventory at event initiation; c_p is its specific heat; ΔT_{sub} is its initial subcooling below saturation; and h_{fg} is its latent heat of vaporization. Everything that follows is an accounting of this balance when the left side — the heat — is aggregated over N modules while the right side — the inventory — may or may not have been sized for N.

2.2 The Single-Module Coping Duration as the Reference Quantity

The design-certification basis establishes, for a single module, that the usable ultimate-heat-sink inventory supports decay-heat rejection for at least the coping duration without makeup, power, or operator action, maintaining the core within its coping design limits and keeping the containment submerged (for immersed-pool designs) or the isolation condenser covered (for isolation-condenser designs) throughout. This single-module duration is the reference quantity. The 72-hour ASSC figure carried in FC1 is best understood as the established regulatory coping benchmark — the post-Fukushima station-blackout and FLEX coping timeline before offsite resources are credited — which the passive designs claim to meet or exceed without operator action; the physical pool-boil duration for some architectures is considerably longer. This distinction is developed in Section 6.

2.3 Why Single-Module Coping Does Not Trivially Extend to N Modules

The intuition that a design proven to cope with one module at a time must cope with N modules at once is false in general, and the reason is precisely the shared substrate that FC1 §2.2 identifies. Where modules share an ultimate-heat-sink inventory, the aggregate heat delivered to that inventory in a simultaneous event is approximately N times the single-module heat, while the inventory is whatever the campus was built with. If that shared inventory was sized only for a single-module event, the simultaneous-N coping duration falls toward the single-module duration divided by N — a potentially disqualifying reduction. If instead the shared inventory was sized for the full-campus aggregate heat load, the certified coping duration already is the simultaneous-N duration, and single-module events are bounded with large margin. The multi-module adequacy question is therefore not a new thermal-hydraulic phenomenon; it is a question about which sizing basis the shared resources were built to. Locating and answering that question is the work of the next two sections.

3. Ultimate-Heat-Sink Architecture Taxonomy

The degree and manner of thermal sharing across modules is set by the ultimate-heat-sink architecture, which differs materially across the near-term integral-LWR families. Three types span the near-term scope. The multi-module aggregation concern takes a different form in each, so the adequacy analysis must first classify the design.

3.1 Type A — Shared Immersed Pool

All modules are immersed in, and reject decay heat to, a single below-grade reactor pool that serves as the common ultimate heat sink for the campus. This is the architecture of the immersed-containment integral-PWR design family. Thermal sharing is maximal: a simultaneous event delivers the aggregate N-module decay heat to one inventory, and the pool level falls under the combined boil-off of all modules at once. For Type A, the governing coupling is aggregate pool boil-off, and adequacy turns directly on whether the certified pool inventory and its licensed coping duration were established on the full-complement simultaneous-actuation basis.

3.2 Type B — Per-Module Isolation-Condenser Pools

Each module rejects decay heat through an isolation condenser or passive containment cooling condenser to its own dedicated pool inventory, sized per module for a multi-day coping duration without makeup. This is the architecture of the natural-circulation boiling-water SMR family. Thermal sharing at the primary heat-rejection stage is low: each module’s coping is largely independent for the without-makeup period. The multi-module concern for Type B is not aggregate boil-off but the shared long-term resources — the common makeup-water supply required to extend coping beyond the per-module pool duration, and any shared ultimate-heat-sink environment into which all condensers ultimately reject. Adequacy for Type B turns on whether the shared makeup provisions were sized for simultaneous refill of N pools.

3.3 Type C — Passive Air-Cooled Containment

Each module rejects decay heat through a passive containment cooling system that transitions, after an initial water-augmented phase, to air cooling of the containment shell, rejecting heat to the atmosphere. This is the architecture of the passively-cooled integral-PWR variant that credits an air-cooled path. Water-inventory sharing is limited to the initial augmentation phase; the long-term sink is the atmosphere, which is not depletable. The multi-module concern for Type C is the adequacy of the initial water-augmentation inventory if shared, and the aggregate thermal and buoyancy interactions of N adjacent air-cooling paths — whether the heated-air plume of a full campus degrades the air-cooling performance any single module was qualified against.

3.4 The Architecture-Dependence of the Aggregation Concern

The three types locate the multi-module concern in different shared resources, summarized below. This classification is the first step of any campus adequacy analysis: it determines which shared resource the bounding analysis of Section 5 must interrogate.

Type	UHS architecture	Governing coupling	Where the aggregation concern lives
A	Single shared immersed reactor pool	Aggregate pool boil-off across all modules	Sizing basis of the shared pool inventory (aggregate vs. per-module)
B	Per-module isolation-condenser pools	Shared makeup supply and shared long-term sink	Sizing of common makeup for simultaneous N-pool refill
C	Passive air-cooled containment	Shared augmentation water; adjacent air-path interactions	Augmentation-inventory sizing; aggregate plume effect on air cooling

4. The Adequacy Decomposition

4.1 Per-Module-Independent Versus Shared-Resource Elements

The campus coping argument decomposes cleanly into two kinds of element. Per-module-independent elements are those whose function for one module does not draw on any resource another module also draws on: the module’s own primary heat-removal path, its own passive actuation logic, its own containment integrity, its own within-module inventory. For these, single-module adequacy extends to the campus without modification — N independent copies of a proven argument. Shared-resource elements are those whose function for one module draws on a resource finite across the campus: the shared ultimate-heat-sink inventory, the shared makeup supply, a shared spent-fuel-pool heat load imposed on the same inventory, a shared air-cooling environment, and shared electrical or pneumatic support to any credited passive-actuation valves.

For these, single-module adequacy does not extend without demonstrating that the shared resource was sized for the aggregate demand.

4.2 The Governing Shared Resources

Five shared resources govern multi-module coping adequacy across the three architectures. First, the ultimate-heat-sink inventory itself, for Type A — the shared pool whose boil-off must not uncover containment before the ASSC duration under aggregate heat. Second, the makeup-water supply, for Type B and for the extended phase of Type A — whether the common source and its delivery can refill N sinks simultaneously without operator action, and whether that source is itself independent of the initiating event. Third, the spent-fuel-pool coupling — where campus spent fuel shares the reactor-pool inventory, its decay-heat load adds to the aggregate and must be carried in the balance rather than analyzed separately. Fourth, the air-cooling transition, for Type A and Type C — the heat load and pool level at which the design transitions from boil-off to air cooling must be reached, for the aggregate load, before inventory is exhausted, and the aggregate air-cooling path must remain qualified. Fifth, shared support to passive-actuation components — any valve, damper, or logic that a passive train credits must not depend on a shared electrical or pneumatic supply that a correlated event removes for all modules at once.

4.3 The Reduction

With the decomposition in hand, the multi-module adequacy question reduces to a single, checkable proposition: the campus sustains the ASSC coping duration under simultaneous N-module actuation if, and only if, each governing shared resource was sized — and its certified coping basis established — for the aggregate N-module demand rather than the single-module demand, with the per-module-independent elements carried unchanged. This reduction is the central analytical result of the supplement. It converts an open-ended thermal-hydraulic worry into a finite list of sizing-basis confirmations, each answerable from design-certification data, and it tells the reviewer exactly which confirmations to demand.

5. The Bounding Adequacy Analysis

5.1 Conservative Assumptions

The adequacy analysis is deterministic and bounding, consistent with design-basis-accident practice. The bounding case assumes: simultaneous actuation of all N modules at the same instant; each module at end-of-cycle peak burnup with the bounding decay-heat curve (ANS-5.1 with uncertainty applied); no operator action; no external AC or DC power beyond that which the passive design credits as always-available; no makeup for the without-makeup coping period; the shared ultimate-heat-sink inventory at its minimum technical-specification level at event initiation; the spent-fuel-pool heat load, where coupled, at its bounding value; a single failure applied to any non-passive component credited in the sequence; and degraded heat-transfer allowances consistent with the design-certification methodology. These assumptions are individually conservative and jointly bounding; the campus that satisfies the acceptance criterion under them satisfies it in any less severe correlated event.

5.2 The Acceptance Criterion

Adequacy is demonstrated when two conditions hold under the bounding case. First, the aggregate coping duration — the time until the usable shared inventory is exhausted or a credited air-cooling transition is established — equals or exceeds the ASSC design-basis duration, with margin consistent with the design-certification acceptance methodology. Second, no cliff-edge is crossed before the ASSC duration: the shared pool level does not fall below the elevation that maintains containment submergence and passive natural circulation (Type A); the isolation-condenser pools do not uncover their condensers before credited makeup (Type B); and the air-cooling path remains within its qualified envelope under the aggregate plume (Type C). A campus that meets the coping-duration condition but crosses a cliff-edge — losing natural circulation as pool level drops, for instance — is not adequate, because coping duration computed on a heat-balance basis is meaningless past the point where the heat-transfer mode it assumed has failed.

5.3 The Calibration-Input Register

The following certified inputs populate the proof for a specific design. Each is design-certification data, carried here as a calibration item; the supplement supplies the structure into which they are placed and the use to which each is put, not the value.

Certified input	Source	Why it governs
Bounding per-module decay-heat curve at EOC peak burnup	Design certification / ANS-5.1 evaluation	Sets the aggregate heat load (×N) that the shared inventory must absorb
Usable shared UHS inventory at minimum tech-spec level	Design certification / technical specifications	Sets the energy the sink can absorb before exhaustion
UHS sizing basis: aggregate (N-module) or per-module	Design certification licensing basis	The single determinant of whether coping scales or collapses with N
Makeup source rate, capacity, and independence	Design certification / FLEX basis	Determines whether coping extends beyond the without-makeup period for N sinks
Spent-fuel-pool heat load and inventory coupling	Design certification	Adds to the aggregate balance where the SFP shares the pool
Air-cooling transition heat load and pool level	Design certification	Sets whether the depletable phase ends before inventory does, for the aggregate

Certified input	Source	Why it governs
Cliff-edge levels (submergence, circulation, condenser cover)	Design certification safety analysis	Bound the coping duration independently of the heat balance

6. The 72-Hour Floor — Reconciliation Finding

6.1 The Class Floor Versus the Family-Certified Durations

The 72-hour figure carried in FC1 is a class floor grounded in the established regulatory coping benchmark — the post-Fukushima station-blackout and FLEX coping timeline. The family-specific certified coping durations are not uniform and, for several architectures, exceed it: large shared-pool designs credit boil-off coping measured in weeks, while isolation-condenser designs credit multi-day per-module durations without makeup. Carrying a single 72-hour class floor is defensible and conservative for a document that spans the class. Still, it understates the margin of the longer-coping families and must not be mistaken for the certified duration of any specific design. FC1 §5A.1 should state the 72-hour figure explicitly as a conservative class floor anchored to the FLEX benchmark, and cite the family-specific certified durations as the governing values for a specific application.

6.2 The Multi-Module Case Is Where the Floor Is Tested

The reconciliation matters most precisely at the multi-module case, because that is where a comfortable single-module margin can be consumed by aggregation. A family that certifies a two-week single-module pool-boil duration has ample margin over the 72-hour floor for one module; whether it retains margin over the floor for a six- or twelve-module simultaneous event depends entirely on the shared-pool sizing basis of Section 4.3. The floor is therefore not tested by the single-module certified duration, however large; it is tested by the aggregate coping duration under the bounding simultaneous case. This is the analytical reason the multi-module question, and not the single-module certified number, is the one that governs the adequacy of the ASSC floor for a campus.

6.3 Recommendation for FC1

Two changes are recommended to FC1 on its next revision pass. First, at §5A.1, frame the 72-hour ASSC duration as a conservative class floor anchored to the FLEX coping benchmark, with the family-specific certified durations named as the governing values, and state that the coping duration credited for any campus is the aggregate simultaneous-actuation duration, not the single-module duration. Second, at §9.1, replace the open calibration item with a reference to this supplement and its reduction (Section 4.3): the residual is closed in method — adequacy is confirmed by verifying that each governing shared resource was sized for the aggregate N-module demand — and remains open in data only until the certified inputs of the calibration register are supplied for a specific design.

7. Relationship to Adjacent Instruments

7.1 To DBA-MA-SMR-FC1 §5A

This supplement substantiates the coping-duration barrier that the passive-safety adequacy result credits. FC1 §5A states the adequacy result and its declared residual; this supplement supplies the thermal-hydraulic basis for the coping duration on which that result depends and the method by which the multi-module residual is closed. The result flows back into FC1 §5A.1 and §9.1 per Section 6.3.

7.2 To DBA-MA-DiD

The correlated-barrier method treats the coping clock as a governing cutset — the persistence-and-denial path that defeats the coping-duration argument by outlasting it. This supplement is the physical substantiation of that cutset: it establishes how long the clock actually runs under aggregate load, which is the quantity the cutset's feasibility screen requires. Where the aggregate coping duration meets the ASSC floor with margin, the clock cutset is bounded in time; where it does not, the shortfall is the physical measure of the residual the correlated-barrier analysis carries forward.

7.3 To the Consequence-Chain Use Case

Where a campus fails the acceptance criterion — a shared resource sized only for the single-module demand — the aggregate coping duration falls short of the ASSC floor, and the degraded-cooling sequence that follows is the initiating condition for the consequence-chain use case (DBA-MA-SMR-UC-1). This supplement thus determines whether the consequence chain is entered at all: adequate shared-resource sizing keeps the campus within the coping envelope; inadequate sizing hands a defined, quantified shortfall to the consequence analysis.

8. Gaps and Open Items

8.1 Certified-Input Population

The calibration register of Section 5.3 is unpopulated pending design-certification data for the specific design families. Populating it — particularly the ultimate-heat-sink sizing basis (aggregate vs. per-module) for each family — is the single step that converts the method-level closure of FC1 §9.1 into a data-level closure for a named design.

8.2 Spent-Fuel-Pool Coupling

Where campus spent fuel shares the reactor-pool inventory, its heat load must be carried in the aggregate balance. The coupled spent-fuel-pool thermal progression is developed in the use-case instrument (DBA-MA-SMR-UC-2); the interface between that use case and this supplement — ensuring the shared inventory is not double-counted or double-committed — is an open coordination item.

8.3 Air-Cooling Transition Under Aggregate Plume

For Type A and Type C architectures, the adequacy of the air-cooling phase under the aggregate heated-air plume of a full campus — as distinct from the single-module air-cooling qualification — requires computational-fluid-dynamics confirmation that is design-specific and not yet available. It is flagged as the least-bounded element of the multi-module analysis.

8.4 Staggered Versus Simultaneous Actuation

The bounding analysis assumes perfectly simultaneous actuation. A real correlated event may actuate modules over a short interval rather than instantaneously; whether staggered actuation meaningfully relieves the aggregate peak, or whether the conservative simultaneous assumption should stand as the design basis, is an open refinement — to be resolved conservatively unless a design-specific analysis justifies credit for stagger.

Revision History

Version	Date	Author	Description
0.1	July 2026	T. Roxey	Initial working draft. Establishes the thermal-hydraulic adequacy framework for the ASSC coping duration under simultaneous multi-module actuation, as a facility-class technical-basis supplement to DBA-MA-SMR-FC1 closing the calibration item at FC1 §9.1. Develops the coping heat balance, stated in integral form for the N-module campus (ANS-5.1 decay-heat basis), the ultimate-heat-sink architecture taxonomy (shared immersed pool / per-module isolation-condenser pools / passive air-cooled containment), the decomposition into per-module-independent and shared-resource elements, and the reduction of multi-module adequacy to shared-resource sizing basis. States the conservative bounding assumptions, the two-part acceptance criterion (coping duration ≥ ASSC floor and no cliff-edge crossing), and the calibration-input register. Records the reconciliation finding that the 72-hour figure is a conservative class floor anchored to the FLEX benchmark, tested — not by the single-module certified duration — but by the aggregate simultaneous-actuation duration, with recommendations to FC1 §5A.1 and §9.1. Quantitative inputs carried as design-certification calibration items. Working document tag DBA-MA-SMR-FC1-TB1 pending confirmation.